

- Of course we do not know exactly what c is. But suppose that we know that the angle was somewhere between 25° and 35° . In other words suppose that, even though we don't know precisely what our measurement error was, it was certainly no more than 5° .
- Now on the range $25^\circ < c < 35^\circ$, $\sec(c)$ is an increasing and positive function. Hence on this range

$$1.217 \dots = \sec^2 25^\circ \leq \sec^2 c \leq \sec^2 35^\circ = 1.490 \dots < 1.491$$

So

$$12.17 \cdot |\Delta\theta| \leq |\Delta h| = 10 \sec^2(c) \cdot |\Delta\theta| \leq 14.91 \cdot |\Delta\theta|$$

- Since we require $|\Delta h| < 0.1$, we need $14.91|\Delta\theta| < 0.1$, that is

$$|\Delta\theta| < \frac{0.1}{14.91} = 0.0067 \dots$$

So we must measure angles with an accuracy of no less than 0.0067 radians — which is

$$\frac{180}{\pi} \cdot 0.0067 = 0.38^\circ.$$

Hence a measurement error of 0.38° or less is acceptable.

Example 3.4.37

3.4.9 ► (Optional) — Derivation of the Error Formulae

In this section we will derive the formula for the error that we gave in equation (3.4.33) — namely

$$R_n(x) = f(x) - T_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c) \cdot (x-a)^{n+1}$$

for some c strictly between a and x , and where $T_n(x)$ is the n^{th} order Taylor polynomial approximation of $f(x)$ about $x = a$:

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a).$$

Recall that we have already proved a special case of this formula for the constant approximation using the Mean-Value Theorem (Theorem 2.13.4). To prove the general case we need the following generalisation⁴⁹ of that theorem:

49 It is not a terribly creative name for the generalisation, but it is an accurate one.

Theorem 3.4.38 (Generalised Mean-Value Theorem).

Let the functions $F(x)$ and $G(x)$ both be defined and continuous on $a \leq x \leq b$ and both be differentiable on $a < x < b$. Furthermore, suppose that $G'(x) \neq 0$ for all $a < x < b$. Then, there is a number c obeying $a < c < b$ such that

$$\frac{F(b) - F(a)}{G(b) - G(a)} = \frac{F'(c)}{G'(c)}$$

Notice that setting $G(x) = x$ recovers the original Mean-Value Theorem. It turns out that this theorem is not too difficult to prove from the MVT using some sneaky algebraic manipulations:

Proof. • First we construct a new function $h(x)$ as a linear combination of $F(x)$ and $G(x)$ so that $h(a) = h(b) = 0$. Some experimentation yields

$$h(x) = [F(b) - F(a)] \cdot [G(x) - G(a)] - [G(b) - G(a)] \cdot [F(x) - F(a)]$$

- Since $h(a) = h(b) = 0$, the Mean-Value theorem (actually Rolle's theorem) tells us that there is a number c obeying $a < c < b$ such that $h'(c) = 0$:

$$\begin{aligned} h'(x) &= [F(b) - F(a)] \cdot G'(x) - [G(b) - G(a)] \cdot F'(x) & \text{so} \\ 0 &= [F(b) - F(a)] \cdot G'(c) - [G(b) - G(a)] \cdot F'(c) \end{aligned}$$

Now move the $G'(c)$ terms to one side and the $F'(c)$ terms to the other:

$$[F(b) - F(a)] \cdot G'(c) = [G(b) - G(a)] \cdot F'(c).$$

- Since we have $G'(x) \neq 0$, we know that $G'(c) \neq 0$. Further the Mean-Value theorem ensures⁵⁰ that $G(a) \neq G(b)$. Hence we can move terms about to get

$$\begin{aligned} [F(b) - F(a)] &= [G(b) - G(a)] \cdot \frac{F'(c)}{G'(c)} \\ \frac{F(b) - F(a)}{G(b) - G(a)} &= \frac{F'(c)}{G'(c)} \end{aligned}$$

as required. □

Armed with the above theorem we can now move on to the proof of the Taylor remainder formula.

Proof of equation (3.4.33). We begin by proving the remainder formula for $n = 1$. That is

$$f(x) - T_1(x) = \frac{1}{2}f''(c) \cdot (x - a)^2$$

50 Otherwise if $G(a) = G(b)$ the MVT tells us that there is some point c between a and b so that $G'(c) = 0$.

- Start by setting

$$F(x) = f(x) - T_1(x) \qquad G(x) = (x - a)^2$$

Notice that, since $T_1(a) = f(a)$ and $T_1'(x) = f'(a)$,

$$\begin{aligned} F(a) &= 0 & G(a) &= 0 \\ F'(x) &= f'(x) - f'(a) & G'(x) &= 2(x - a) \end{aligned}$$

- Now apply the generalised MVT with $b = x$: there exists a point q between a and x such that

$$\begin{aligned} \frac{F(x) - F(a)}{G(x) - G(a)} &= \frac{F'(q)}{G'(q)} \\ \frac{F(x) - 0}{G(x) - 0} &= \frac{f'(q) - f'(a)}{2(q - a)} \\ 2 \cdot \frac{F(x)}{G(x)} &= \frac{f'(q) - f'(a)}{q - a} \end{aligned}$$

- Consider the right-hand side of the above equation and set $g(x) = f'(x)$. Then we have the term $\frac{g(q) - g(a)}{q - a}$ — this is exactly the form needed to apply the MVT. So now apply the standard MVT to the right-hand side of the above equation — there is some c between q and a so that

$$\frac{f'(q) - f'(a)}{q - a} = \frac{g(q) - g(a)}{q - a} = g'(c) = f''(c)$$

Notice that here we have assumed that $f''(x)$ exists.

- Putting this together we have that

$$\begin{aligned} 2 \cdot \frac{F(x)}{G(x)} &= \frac{f'(q) - f'(a)}{q - a} = f''(c) \\ 2 \frac{f(x) - T_1(x)}{(x - a)^2} &= f''(c) \\ f(x) - T_1(x) &= \frac{1}{2!} f''(c) \cdot (x - a)^2 \end{aligned}$$

as required.

Oof! We have now proved the cases $n = 1$ (and we did $n = 0$ earlier).

To proceed — assume we have proved our result for $n = 1, 2, \dots, k$. We realise that we haven't done this yet, but bear with us. Using that assumption we will prove the result is true for $n = k + 1$. Once we have done that, then

- we have proved the result is true for $n = 1$, and
- we have shown if the result is true for $n = k$ then it is true for $n = k + 1$

Hence it must be true for all $n \geq 1$. This style of proof is called mathematical induction. You can think of the process as something like climbing a ladder:

- prove that you can get onto the ladder (the result is true for $n = 1$), and
- if I can stand on the current rung, then I can step up to the next rung (if the result is true for $n = k$ then it is also true for $n = k + 1$)

Hence I can climb as high as like.

- Let $k > 0$ and assume we have proved

$$f(x) - T_k(x) = \frac{1}{(k+1)!} f^{(k+1)}(c) \cdot (x-a)^{k+1}$$

for some c between a and x .

- Now set

$$F(x) = f(x) - T_{k+1}(x) \quad G(x) = (x-a)^{k+1}$$

and notice that, since $T_{k+1}(a) = f(a)$,

$$F(a) = f(a) - T_{k+1}(a) = 0 \quad G(a) = 0 \quad G'(x) = (k+1)(x-a)^k$$

and apply the generalised MVT with $b = x$: hence there exists a q between a and x so that

$$\begin{aligned} \frac{F(x) - F(a)}{G(x) - G(a)} &= \frac{F'(q)}{G'(q)} && \text{which becomes} \\ \frac{F(x)}{(x-a)^{k+1}} &= \frac{F'(q)}{(k+1)(q-a)^k} && \text{rearrange} \\ F(x) &= \frac{(x-a)^{k+1}}{(k+1)(q-a)^k} \cdot F'(q) \end{aligned}$$

- We now examine $F'(q)$. First carefully differentiate $F(x)$:

$$\begin{aligned} F'(x) &= \frac{d}{dx} \left[f(x) - \left(f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 + \cdots + \frac{1}{k!}f^{(k)}(a)(x-a)^k \right) \right] \\ &= f'(x) - \left(f'(a) + \frac{2}{2}f''(a)(x-a) + \frac{3}{3!}f'''(a)(x-a)^2 + \cdots + \frac{k}{k!}f^{(k)}(a)(x-a)^{k-1} \right) \\ &= f'(x) - \left(f'(a) + f''(a)(x-a) + \frac{1}{2}f'''(a)(x-a)^2 + \cdots + \frac{1}{(k-1)!}f^{(k)}(a)(x-a)^{k-1} \right) \end{aligned}$$

Now notice that if we set $f'(x) = g(x)$ then this becomes

$$F'(x) = g(x) - \left(g(a) + g'(a)(x-a) + \frac{1}{2}g''(a)(x-a)^2 + \cdots + \frac{1}{(k-1)!}g^{(k-1)}(a)(x-a)^{k-1} \right)$$

So $F'(x)$ is then exactly the remainder formula but for an order $k-1$ approximation to the function $g(x) = f'(x)$.